As described at the beginning of the section, when using the path compression heuristic, we don't update the root of all elements on `merge`, but we do update it on `findPartition`. So, the main difference with the older version is that we save the result of the recursive calls to `findPartition` at line 5, and we use it to update current element's root. Everything else, remains exactly the same.

It goes without saying that the largest portion of changes will be implemented in the `merge` method, shown in listing 5.10.

Listing 5.10 Tree-with-heuristics solution, merge

```
function merge(elem1,
elem2)                                          #1
  r1 = this.findPartition(elem1)
                                  #2
  r2 =
this.findPartition(elem2)
#3
  if r1 == r2
then                                                  #4
    return false
  info1 = this.parentsMap[r1]
                                       #5
  info2 = this.parentsMap[r2]

  if info1.rank >= info2.rank
then                                   #6
info2.root = info1.root;
                                            #7
info1.rank += info2.rank;
                                         #8
  else
info1.root =
info2.root;                                            #9
```